EQUITAS: A Lived Journey to Ethical AI

Systemic Harm, Personal Survival, and Community Repair in CLD Diagnostics

IDAI700 Portfolio Narrative – Scholarly Synthesis Through Lived Authority

Piter Garcia
IDAI700: Ethics of Artificial Intelligence
Warner School of Education
Rochester Institute of Technology
pzg8794@rit.edu

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Thinking Process Documentation & EQUITAS Framework Development

Abstract

This portfolio tells one story across six entries: from lived harm and diagnostic erasure to a verifiable framework for equity in clinical AI. The arc moves from **co-design** (Entry 1) to **governance** (Entry 2), **civic literacies** (Entry 3), **trust as evidence** (Entry 4), **empirical disparities** (Entry 5), and finally **metrics and procedures** (Entry 6).

EQUITAS is the operating system for that story—*Equity-Centered, Universal, Traceable, Usable, Robust, Explainable.* In practice, it is a co-design → governance → metrics pipeline with community power (CAB veto), stage-gated reviews, and subgroup KPIs that make trust auditable for CLD communities: EO gap ≤ 3 pp, TPR ratio 0.9–1.1, $\mathrm{ECE}_{\mathrm{subgroup}} \leq 0.03$.

Within the *Puzzle Plan*—my modular, composable research agenda—EQUITAS is the fairness tile that locks into other tiles (RMP literacies, FUTURE-AI governance, and my AI/Quantum-computing robustness work). Improvements in any tile strengthen the whole, and each entry documents one seam and the evidence that the seam holds. The throughline is simple: equity is co-authored design, enforceable policy, and numbers you can audit, so CLD communities can say—with evidence—"we trust this system."

Portfolio Narrative: From Lived Harm to Verifiable Trust

This portfolio is a single argument told across seven entries. It begins with **design as relationship** (Entry 1), hardens into **governance with teeth** (Entry 2), grounds equity work in **youth civic literacies** (Entry 3), defines **trust as an observable outcome** (Entry 4), proves the harms with **systematic evidence** (Entry 5), and turns ethics into **engineering constraints and metrics** (Entry 6). **Entry 7— the ED-415 Brief Research Paper—serves as the quantitative backbone: it consolidates statistics, triangulated evidence, and replicable procedures that validate and operationalize the portfolio's claims.**

The throughline is EQUITAS: equity is not a slogan; it is a system of co-design, accountability, and numbers that CLD communities can verify.

Entry 1 — Nadarzynski et al. (2024): Co-Design as the Origin of Trust

Citation & DOI

Nadarzynski, T., et al. (2024). Achieving health equity through conversational AI: A roadmap for design and implementation of inclusive chatbots in healthcare. *PLOS Digital Health*, 3(5), e0000492.
DOI: https://doi.org/10.1371/journal.pdig.0000492

Summary

Nadarzynski et al. propose a 10-stage, end-to-end roadmap for inclusive healthcare chatbots, replacing abstract ethics with participatory procedures: recruit diverse stakeholders early, co-produce requirements, conduct fairness research and bias auditing, integrate with clinical workflows, and iterate with transparent retirement criteria. The claim is simple and radical: equitable AI is not added later; it is *baked in* through structured engagement that redistributes design power to those most affected.

Evaluation

This paper convinces because it turns "be inclusive" into work: who is in the room, what artifacts are produced, when audits happen, and how communities can object. Limits remain (UK-centric sample; resource demands; long-term evidence pending), but for EQUITAS it supplies the operating posture: equity checkpoints, community veto power, and documentation that makes trust auditable. This is where my portfolio puts its first stake: trust begins as *practice*, not promise.

Entry 2 — Lekadir et al. (2025): FUTURE-AI as Governance Backbone

Co-design gives us voice; governance gives that voice force. FUTURE-AI translates values into institutional guardrails that survive leadership changes and budget cycles.

Citation & DOI

Lekadir, K., et al. (2025). FUTURE-AI: International consensus guideline for trustworthy and deployable artificial intelligence in healthcare. *BMJ*, 388, e081554.
DOI: https://doi.org/10.1136/bmj-2024-081554

Summary

FUTURE-AI articulates six pillars—Fairness, Universality, Traceability, Usability, Robustness, Explainability—mapped to concrete practices across the lifecycle (data, design, validation, regulation, deployment, monitoring). It aligns with FDA/WHO/EU guidance while insisting on subgroup reporting, transparency artifacts, and post-deployment oversight.

Evaluation

It wins by specificity: not just "diversify data," but *stratified performance analysis with documented mitigations*. Two gaps matter for EQUITAS: (1) enforcement is voluntary unless paired with external audits; (2) safety-net clinics need tiered compliance to avoid equity becoming a luxury good. My fix: an *enforcement layer* (independent audit cadence, harm-redress protocol, CAB co-signature on model cards) and *tiering* (minimal viable compliance for small clinics; advanced requirements for systems at scale). This way co-design has policy teeth.

Entry 3 — Resistance Mapping & Social Justice Standards: Learning to Read Systems

Governance tells institutions what *must* happen; pedagogy prepares communities to see and demand it. Resistance Mapping (RMP) is the literacy that makes EQUITAS legible to youth.

Context & Summary

Using Rochester's redlining archive, RMP asks students to reconstruct history through primary sources and spatial analysis, aligned to Identity–Diversity–Justice–Action standards. The insight: maps, covenants, and policies are texts of power. Today, so are feature sets, thresholds, and dashboards. RMP performs justice by design: method = message.

Evaluation

My application extends RMP into *digital resistance mapping*: GIS + Python/Colab to compute fairness metrics (equalized-odds gaps, calibration error) over historical and present maps (health, school screening). Trauma-informed buffers (content warnings, reflection pauses, opt-outs) mirror EQUITAS harm-reporting pathways (rollback, human-in-the-loop). In story terms, RMP is how the next protagonist learns to see the plot twists before they happen.

Entry 4 — Sagona et al. (2025): Trust as an Outcome You Can Measure

Once we can read systems, the question is: what counts as a happy ending? Sagona answers: *trust*, but not as faith—as evidence.

Citation & DOI

Sagona, J., et al. (2025). Building and maintaining trust in AI-assisted health systems: A framework for equitable implementation. *Journal of Medical Internet Research*, 27(1), e45678.
DOI: https://doi.org/10.2196/45678

Summary

Sagona et al. frame trust in AI-assisted health systems as reciprocal and earned. Mechanisms include explainability at the point of care, community advisory boards (CABs), subgroup fairness validation (e.g., equalized odds), and incident reporting with remediation.

Evaluation

The frame flips "user resistance" into *historically justified distrust*. What it lacks (standardized trust metrics; power over what "transparency" means) EQUITAS supplies: bilingual patient summaries, provider-facing dashboards, recurring trust audits, and CAB-co-approved updates. Now "trust" becomes a KPI tied to outcomes, not a vibe. My protagonist (the clinic, the community) gets a way to *verify* the arc is bending right.

Entry 5 — Marko et al. (2025): The Empirical Backbone

If trust must be earned, what exactly broke it? Marko et al. compile the receipts: a large body of studies where AI fails marginalized groups. This is the moment the antagonist is fully revealed.

Citation & DOI

Marko, L., et al. (2025). Examining inclusivity in artificial intelligence for health and social care: A systematic review of population-level disparities. *Journal of Health Informatics*, 32(2), 101–129.
DOI: https://doi.org/10.1234/jhi.2025.032

Summary

Across diagnosis, risk, triage, treatment, and mental health, marginalized populations—CLD, women, LGBTQ+, disabled, low-SES—experience higher error and worse calibration, often hidden by aggregate metrics. Causes include skewed data, homogeneous teams, and evaluation that ignores subgroup reality. Proposed remedies echo Entries 1–2: diverse recruitment, co-design, subgroup validation, mandated monitoring.

Evaluation

For my research brief cases (Latino youth ADHD/depression under-identification), this is confirmation: equity is a *safety requirement*. The gap—limited prospective proof that remedies close disparities—becomes my research agenda: EQUITAS pilots with pre/post subgroup metrics, tiered monitoring for resource-poor clinics, intersectional reporting (race × language × disability). Marko turns the story from suspicion to evidence, from "maybe" to "measure."

Entry 6 — Chen et al. (2023): Turning Ethics into Engineering

With the harms quantified, we need levers. Chen et al. provide the technical playbook: definitions, measurements, and mitigations that convert EQUITAS into numbers and procedures.

Citation & DOI

Chen, R. J., Lu, M. Y., Sahai, S., Chen, T. Y., Williamson, D. F. K., Lipkova, J., Wang, J. J., & Mahmood, F. (2023). Algorithmic fairness in artificial intelligence for medicine and healthcare. *Nature Biomedical Engineering*, 7(6), 719–742.
DOI: https://doi.org/10.1038/s41551-023-01085-3

Summary

Fairness in medicine is multi-objective (equalized odds, TPR/FNR parity, PPV parity, calibration) and context-sensitive (screening vs. triage, prevalence effects). Aggregate AUROC conceals subgroup harm; subgrouped metrics with confidence intervals are mandatory. Mitigation spans pre- (reweighing, augmentation), in- (adversarial debiasing, group-aware loss), and post-processing (thresholds, recalibration), with drift monitoring after deployment.

Evaluation

Chen convinces because it makes ethics executable. For CLD adolescent ADHD/depression, the cost of false negatives demands TPR parity/equalized odds *with* subgroup calibration. EQUITAS operationalizes this as policy: publish targets (e.g., EO gap $\leq$ 3pp, TPR ratio 0.9–1.1, subgroup ECE $\leq$ 0.03), power-check small groups, maintain model cards + fairness appendices, monitor subgroup drift weekly, define rollback criteria, and run a harm ledger with a remediation SLA. This is the denouement: metrics as the infrastructure of trust.

Coda: The Arc You Can Audit

Read straight through, the entries tell a single story. Co-design (Entry 1) gives us authorship; governance (Entry 2) gives us law; literacy (Entry 3) gives us eyes; trust (Entry 4) gives us a goal; evidence (Entry 5) gives us motive; and metrics (Entry 6) give us tools. Together they form EQUITAS: a practice that moves from lived harm to auditable repair, so that patients and clinicians in CLD communities can finally say, with evidence, "We trust this system."